Example 5.19 Calculate the slope and y intercept of the straight line $8x - 7y + 6 = 0$

Solution Equation of the given straight line is $8x - 7y + 6 = 0$

$7y = 8x + 6$ (bringing it to the form $y = mx + c$)

$$y = \frac{8}{7}x + \frac{6}{7} \quad \ldots(1)$$

Comparing (1) with $y = mx + c$

Slope $m = \frac{8}{7}$ and y intercept $c = \frac{6}{7}$

DO YOU KNOW?

For, the point (x, y) in a xy plane, the x coordinate x is called "Abscissae" and the y coordinate y is called "Ordinate".

Example 5.20 The graph relates temperatures y (in Fahrenheit degree) to temperatures x (in Celsius degree) (a) Find the slope and y intercept (b) Write an equation of the line (c) What is the mean temperature of the earth in Fahrenheit degree if its mean temperature is $25°$ Celsius?

Solution (a) From the figure, slope $= \frac{\text{change in } y \text{ coordinate}}{\text{change is } x \text{ coordinate}} = \frac{68 - 32}{20 - 0} = \frac{36}{20} = \frac{9}{5} = 1.8$

The line crosses the Y axis at $(0, 32)$

So the slope is $\frac{9}{5}$ and y intercept is 32.

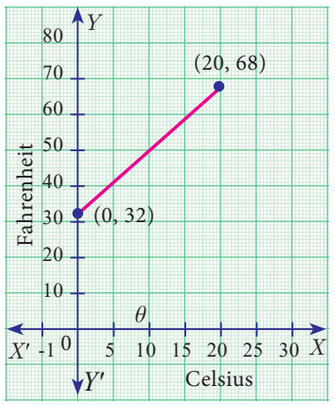


Fig. 5.31

(b) Use the slope and y intercept to write an equation

The equation is $y = \frac{9}{5}x + 32$

(c) In Celsius, the mean temperature of the earth is $25°$. To find the mean temperature in Fahrenheit, we find the value of y when $x = 25$

$$y = \frac{9}{5}x + 32$$

$$y = \frac{9}{5}(25) + 32$$

$$y = 77$$

Note

The formula for converting Celsius to Fahrenheit is given by $F = \frac{9}{5}C + 32$. which is the linear equation representing a stragiht line derived in the example.

Therefore, the mean temperature of the earth is $77°$ F.

5.5.5 Point-Slope form

Here we will find the equation of a straight line passing through a given point $A(x_1, y_1)$ and having the slope m.

Let $P(x, y)$ be any point other than A on the given line. Slope of the line joining $A(x_1, y_1)$ and $P(x, y)$ is given by

$$m = \tan\theta = \frac{y - y_1}{x - x_1}$$

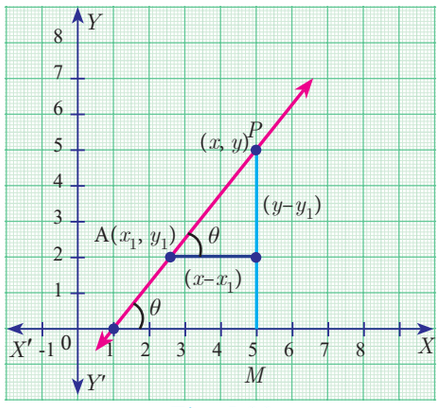


Fig. 5.32

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